

HEART STRUCTURE

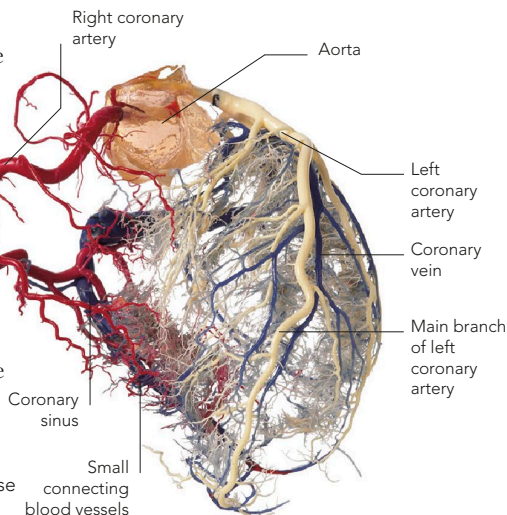
THE HEART IS A POWERFUL ORGAN ABOUT THE SIZE OF A CLENCHED FIST. LOCATED JUST TO THE LEFT OF CENTRE IN BETWEEN THE LUNGS, IT OPERATES AS TWO COORDINATED PUMPS THAT SEND BLOOD AROUND THE BODY.

THE HEART'S BLOOD SUPPLY

The muscular wall, or myocardium, of the heart is constantly active and needs a generous supply of oxygen and energy from blood. To provide this, the heart muscle has its own blood vessels – the right and left coronary arteries. These vessels branch from the main artery, the aorta, just after it leaves the heart, and send smaller blood vessels into the heart muscle. Waste from heart tissue is removed by the coronary veins, in particular by the coronary sinus, a large vein at the back of the heart.

CORONARY VESSELS

There are many connecting vessels between the coronary arteries. If an artery becomes blocked, these can provide an alternative route for the blood flow.



CARDIAC SKELETON

In the upper heart, four rigid, cuff-like rings, known as the cardiac skeleton, provide points of attachment for the four heart valves and for the heart muscle. The wrap-around muscle fibres in the ventricle walls, and the timing of their contractions, means the ventricles squirt blood from the apex (lower end) upwards, and out through the pulmonary and aortic valves, rather than squeezing blood down to pool in the apex region.

FIBROUS FRAMEWORK

Four rings of fibrous tissue in the heart are known as the cardiac skeleton. Their rigidity prevents the valves from deforming.

